

GemmaLink Development Roadmap

Target: Gemma 4 dev.to Challenge | Dev: Senior Software Architect

This roadmap details the tasks required to build a **Zero-Footprint** Go-based orchestrator for the Gemma 4 multimodal model, designed for non-technical users to access local AI power via their smartphones.

1. Project Setup

Initialize Go workspace and **Wails** desktop framework.

Structure the `/bin` directory for cross-platform sidecar binaries (Windows/Linux).

2. Orchestration Layer

Implement subprocess management in Go (`os/exec`) for the inference engine.

Build the automated GGUF model downloader for 8GB/16GB configurations.

3. LAN Web Interface

Network IP discovery and QR code generation for mobile pairing.

Lightweight Go HTTP server for serving the mobile "Lens" interface.

4. Multimodal Integration

Connect JS MediaDevices API to the Go backend for audio/image streaming.

Prompt engineering for the Gemma 4 **128K context window** and reasoning modes.

5. Deployment & Testing

Cross-compile for WSL Debian (Linux) and Windows 11.

Finalize the single-executable installer bundle.

Deadline: May 24, 2026